Active sensing explains cortical variability and improves visual information

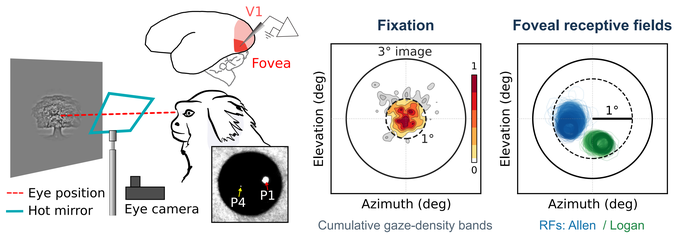
Declan P. Rowley[2], Ryan A. Ressmeyer[1], Tejas Kothapalli[3,4], Alexander C. Huk[2], Jude F. Mitchell[5], Daniel A. Butts[6], Jacob L. Yates[3,4]

[1]Department of Bioengineering, University of Washington; [2]Fuster Laboratory for Cognitive Neuroscience, Departments of Psychiatry & Biobehavioral Sciences and Ophthalmology, UCLA
[3]Redwood Center for Theoretical Neuroscience, UC Berkeley; [4]Herbert Wertheim School of Optometry and Vision Science, UC Berkeley
[5]Brain and Cognitive Science, University of Rochester; [6]Department of Biology, University of Maryland, College Park

UCLA Stein Eye Institute · UC Berkeley

Read the preprint

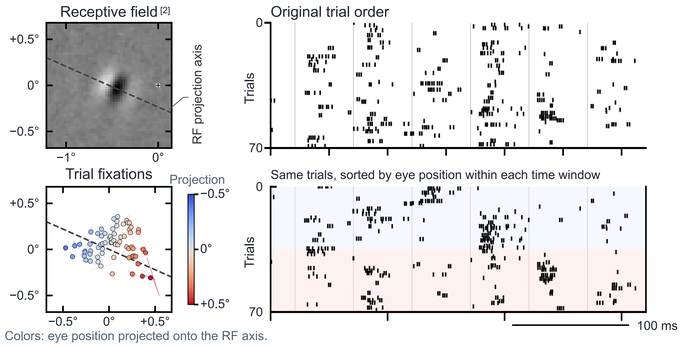
1 Foveal V1 recordings from fixating marmosets



2	1,022	19	20 Hz
awake marmosets	foveal V1 units	recording sessions	repeated natural images

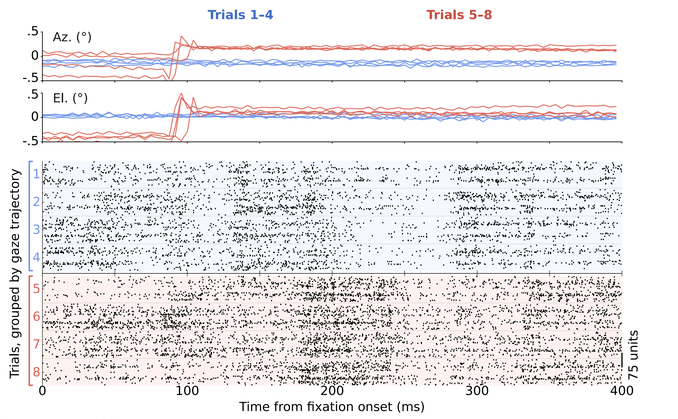
Eye tracking: digital dual-Purkinje imaging (P1 and P4 reflections) [2–4]

2 Small eye movements structure foveal V1 activity



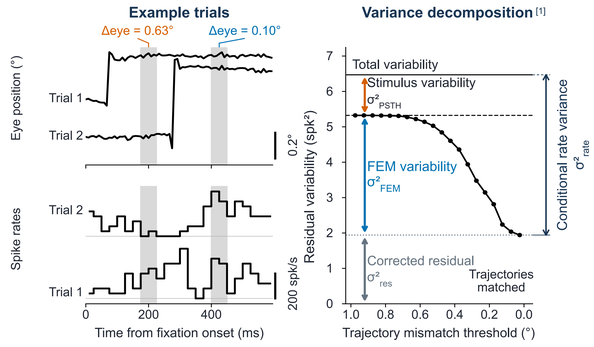
Colors: eye position projected onto the RF axis.

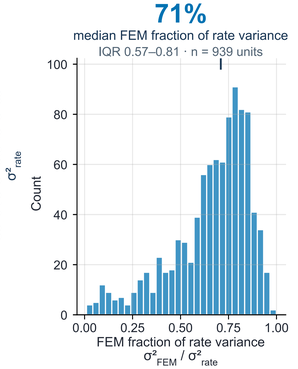
3 Similar gaze trajectories produce similar population responses



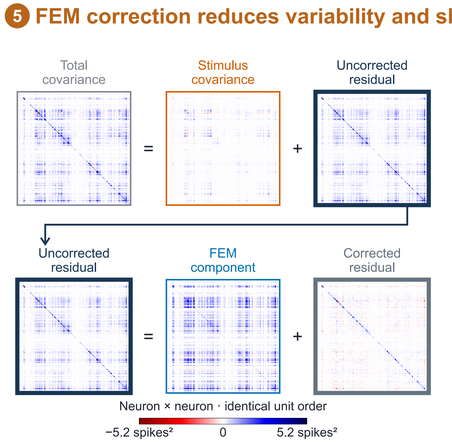
Same stimulus · 132 units.

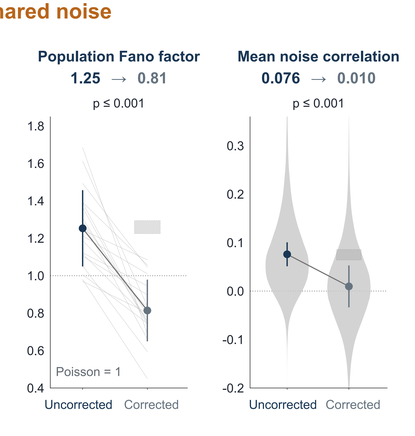
4 FEMs dominate conditional rate variance





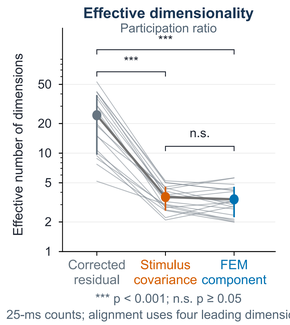
5 FEM correction reduces variability and shared noise

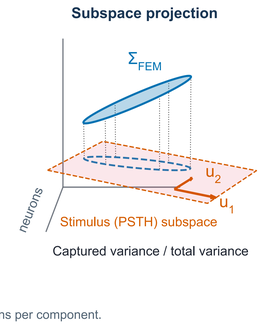


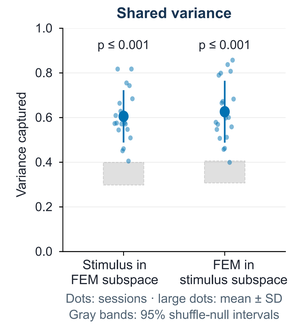


19 sessions; 25-ms bins. Points: mean ± SD. Gray bands: 95% shuffle null; violins: pairwise correlations.
Corrected mean noise correlation: 95% CI −0.010 to 0.029. Both tests compare correction with shuffle.

6 FEM covariance is compact and aligned with stimulus-driven dimensions

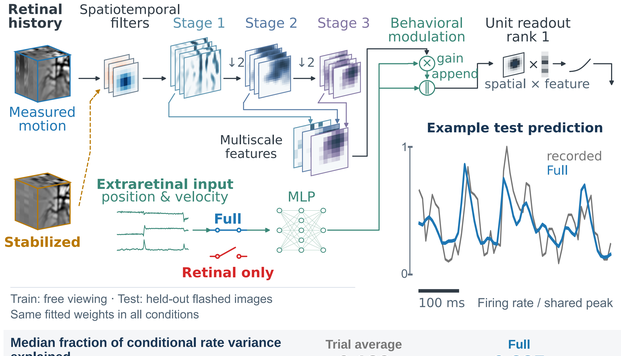


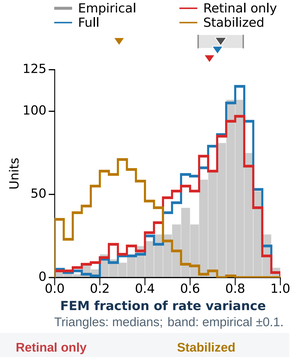




25-ms counts; alignment uses four leading dimensions per component.
Dots: sessions · large dots: mean ± SD
Gray bands: 95% shuffle-null intervals

7 Gaze-dependent retinal input predicts single-trial responses

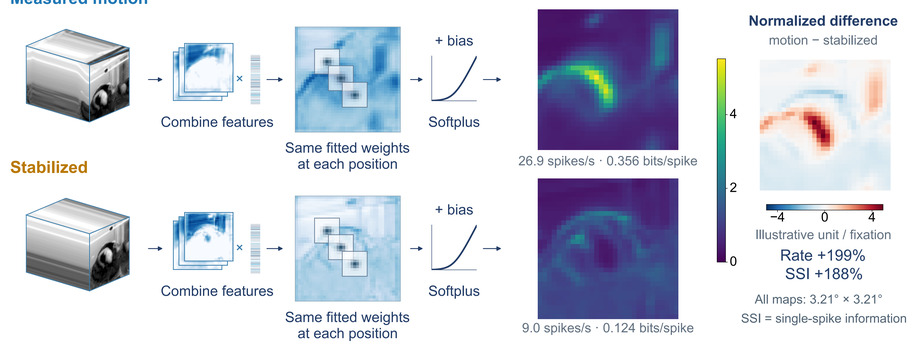




Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions
Triangles: medians; band: empirical ±0.1.

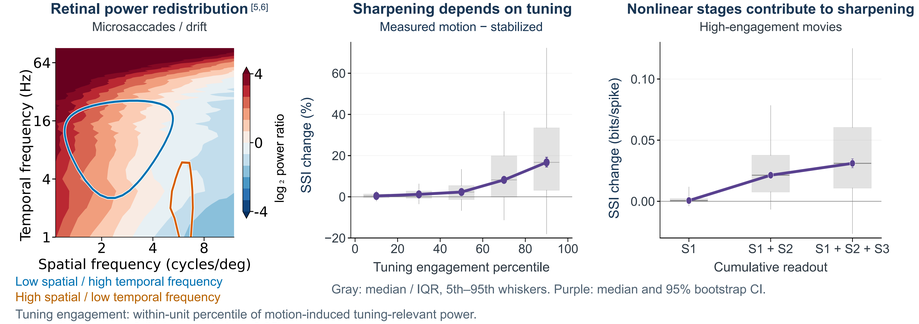
Median fraction of conditional rate variance explained 962 units · 19 sessions	Trial average	Full	Retinal only	Stabilized
	0.139	0.235	0.226	0.042

8 Fixational eye movements sharpen spatial coding in the model



9 Fixational motion engages tuning and sharpens spatial coding

725 units



Tuning engagement: within-unit percentile of motion-induced tuning-relevant power.
All quintiles/stages > 0 (Holm-adjusted p < 0.05).
Gray: median / IQR, 5th–95th whiskers. Purple: median and 95% bootstrap CI.

References
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2025 · [5] Rucci & Victor 2015 · [6] Kuang 2012

Funding
NEI K99-EY037839-01 (D.P.R.); R01-EY030998 (J.F.M.); R01-EY018363 (M.R.).
Fuster Endowment (A.C.H.); NEI K99/R00-EY032179 (J.L.Y.).

Contact
Declan P. Rowley · UCLA
declanrowley@g.ucla.edu